

Name _____ Date _____ Period _____

Transformations of $f(x) = |x|$, Part II

Algebra II | Unit 2 | Day 3 | TEK A2.6.C

Objective: I can describe and graph the effect of c and d on $f(x) = a|x-c|+d$, and combine all four transformations into one function.

Vocabulary

Horizontal shift — $f(x-c)$ moves the graph left/right. $c>0$ shifts _____, $c<0$ shifts _____ - opposite the sign.

Vertical shift — $f(x)+d$ moves the graph up/down. $d>0$ shifts _____, $d<0$ shifts _____ - same as the sign.

Vertex, general form — for $f(x) = a|x-c|+d$, the vertex is always at _____.

Part 1 — Horizontal Shift: $f(x - c)$

Function	Shift	Vertex
$f(x) = x - 3 $	_____	_____
$f(x) = x + 2 $	_____	_____
$f(x) = x - 7 $	_____	_____

Part 2 — Vertical Shift: $f(x) + d$

Function	Shift	Vertex
$f(x) = x + 4$	_____	_____
$f(x) = x - 5$	_____	_____
$f(x) = x + 1.5$	_____	_____

Part 3 — Combine All Four: $f(x) = a|x - c| + d$

"Compressed by $1/3$, shifted down 5, shifted left 2" becomes one function:

$f(x) =$ _____ vertex: _____

You Try. Describe $f(x) = -2|x - 1| + 6$, and give its vertex.

_____ vertex: _____

"Stretched by 4, shifted up 2, shifted right 5" becomes one function:

$f(x) =$ _____ vertex: _____

Part 4 — Real World: Dunhill Drafting's Roofline

Roofline data was modeled by trial-and-error transformation, landing on $f(x) = -1.2|x - 3| + 8$.

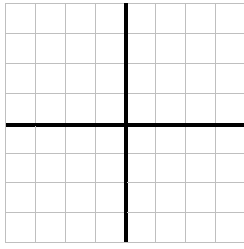
Vertex (highest point of the roof): _____

Describe the transformations from the parent:

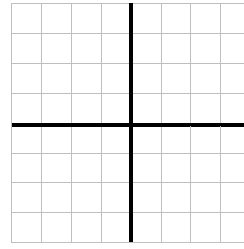
Part 5 — Graph and Compare

Graph both. Each square is 1 unit; both axes run -5 to 5.

A. $f(x) = |x - 2| - 3$



B. $f(x) = |x + 3| + 2$



Vertex of A: _____ Vertex of B: _____

Practice

1. Give the vertex of $f(x) = |x + 5| - 1$. _____
2. Give the vertex of $f(x) = 4|x - 6| + 2$. _____
3. Write a function with vertex $(-4, 7)$ and no stretch/reflection. _____
4. Give the vertex of $f(x) = -3|x + 8| - 6$. _____
5. Write a function with vertex $(2, -9)$ and a vertical stretch by 5. _____
6. Give the vertex of $f(x) = (1/4)|x - 10| + 3$. _____
7. A function has vertex $(-6, 0)$ and opens downward with no stretch. Write it. _____
8. True or False: every function of the form $a|x - c| + d$ has the SAME shape as the parent, just moved and resized. Explain. _____
9. Give the vertex of $f(x) = 2|x + 1| - 4$, and state whether it opens up or down. _____
10. Write a function with vertex $(0, -3)$ that is a reflection of the parent with no stretch.

Exit Ticket

Describe every transformation of $f(x) = -(1/2)|x + 3| - 4$ from the parent, and give its vertex.

_____ vertex _____